High Time-Resolution Queue Profile Estimation at Signalized Intersections Based on Extended Kalman Filtering

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TRB Annual Meeting 2022 Number: TRBAM-22-02414

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Introduction

➢ Motivation

The dynamic spatiotemporal characteristics of queues at urban intersections are crucial to several traffic operation tasks such as signal performance measures, signal optimization and eco-driving vehicle trajectory optimization. Motivated by three facts as follows, **This work combine, for the first time, the shockwave theory with real-time CV data via extending Kalman filtering (EKF) for second-by-second urban queue estimation.**

- Connected vehicles (CVs) are providing unprecedented opportunities for significantly improved traffic state estimation.
- High time-resolution (i.e., every one or a few seconds) queue estimation is a fundamental requirement for more advanced traffic signal control methods, such as max-pressure or SURTRAC algorithms.
- Kalman filtering (KF) and extended Kalman filtering (EKF) have been widely used in traffic state estimation with satisfactory results, especially on freeways.

➢ Challenge

Filtering algorithm is composed of two key terms: model term and measurement term.

- Firstly, the model employed in the filtering framework is dynamic in nature, the shockwave theory however does not fit this requirement.
- Secondly, state of shockwave is not directly available via CV data, especially when market penetration rate (MPR) of CVs is unknown.

Methodology

To estimate the queue profile, QFW and QDW are tracked separately through the EKF in this paper. The EKF is composed of two key terms: the model term and the measurement term. In this paper, a machine-learning-based regression approach is used to construct the model term, and the measurement term is developed by shockwave theory based on real-time CVs data. The overall queue estimation algorithm is illustrated as follows.

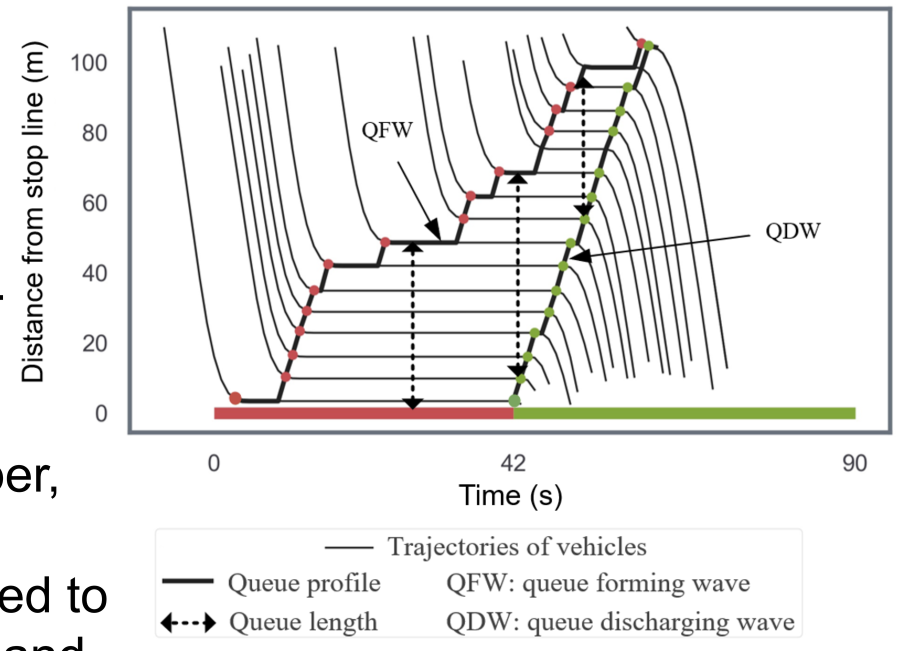

Fig. 1: Illustration of terminologies in space-time diagram

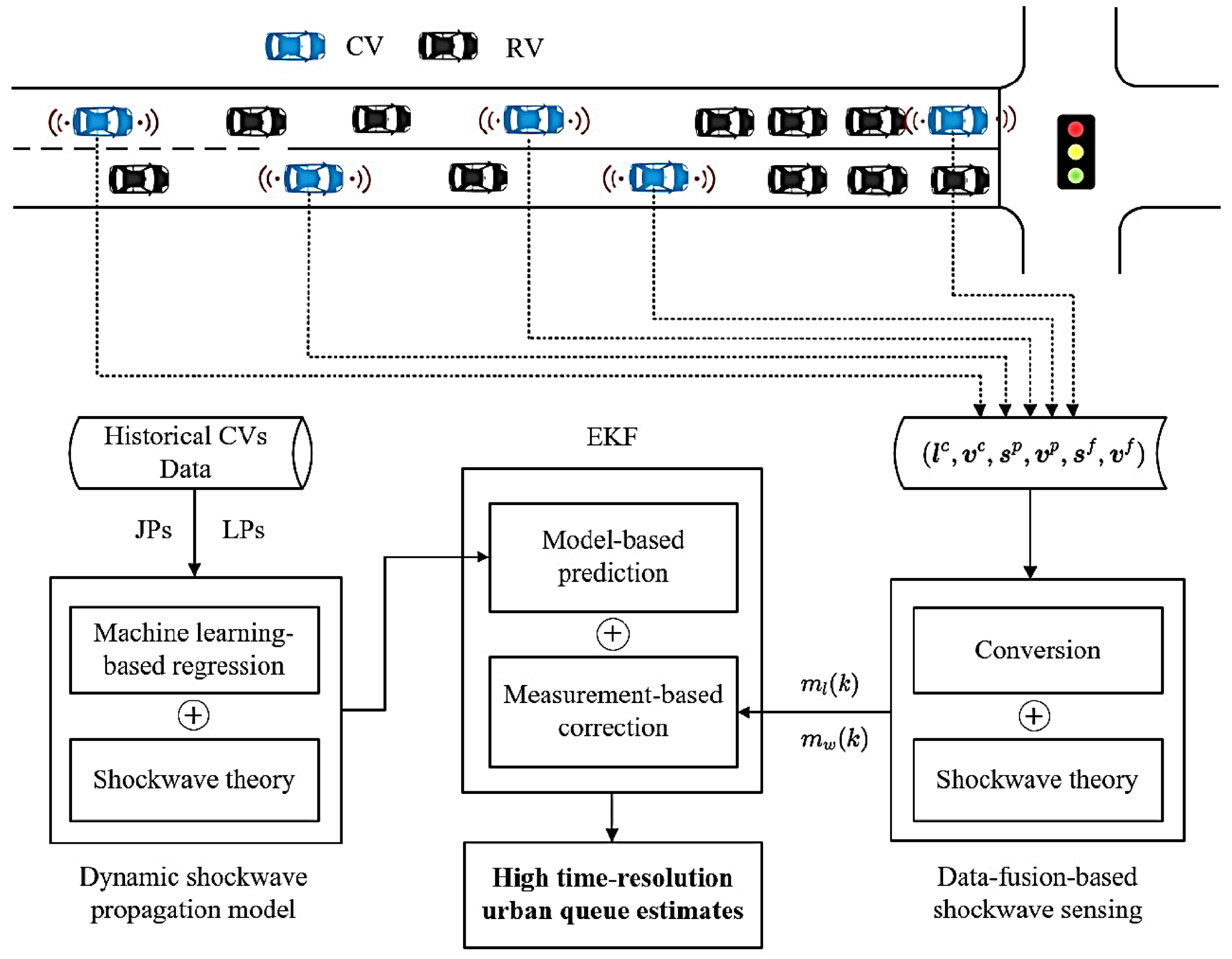

Fig. 2: Traffic measurements and queue profile estimation based on shockwave theory and extended Kalman filtering

Numerical Experiments and Conclusion

➢ Case study with simulation data

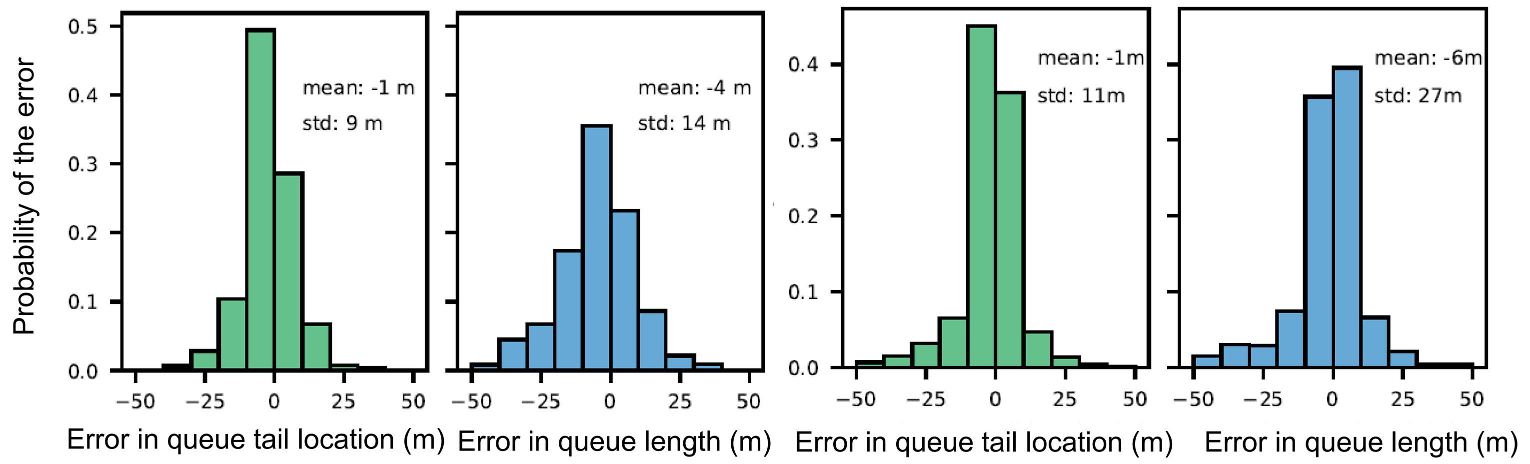

Fig. 3: Histograms for errors of queue profile estimation in 20% penetration rate under different traffic condition: left: under-saturated; lower: over-saturated.

- **Under-saturated:** Around 75% of the time the proposed algorithm provides queue tail estimation with an error from -10 m to 10 m. More than 60% of the time the proposed algorithm provides queue length estimation with an error from -10 m to 10 m.
- **Over-saturated:** Around 80% of the time the proposed algorithm provides queue tail estimation with an error from -10 m to 10 m. More than 70% of the time the proposed algorithm provides queue length estimation with an error from -10 m to 10 m.

Table. 1: MAE and RMSE for estimated queue tail location and queue length under simulation environment

MPR	over-saturated				under-saturated			
	Queue tail location error		Queue length error		Queue tail location error		Queue length error	
	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)
0.05	19.4	40.77	12.11	17.91	12.46	18.24	10.66	14.36
0.1	13.19	28.57	10.01	15.56	10.32	15.05	9.14	12.74
0.2	6.66	15.43	6.51	11.55	5.74	10.09	5.63	9.11
0.3	3.62	9.09	4.04	7.44	4.5	9.53	4.47	8.36
0.4	2.02	5.88	2.78	5.6	2.79	6.63	2.93	6.16
0.5	1.53	4.97	2.26	4.57	1.88	4.82	2.18	4.86
0.6	0.88	3.08	1.69	3.28	1.57	4.67	1.84	4.69
0.7	0.49	1.78	1.36	2.54	0.78	2.56	1.08	2.75
0.8	0.41	1.52	1.28	2.38	0.51	1.69	0.83	2.04
0.9	0.33	1.24	1.2	2.21	0.45	1.62	0.78	1.99
1	0.29	0.99	1.16	2.08	0.45	1.64	0.78	2.01

➢ Case study with real-world data

Table. 2: MAE and RMSE for the queue tail location and queue length under field test.

MPR	field survey				
	Queue tail location error		Queue length error		
	MAE (m)	RMSE (m)	MAE (m)	RMSE (m)	
0.05	14.24	19.78	12.99	17.39	2-3 veh
0.1	12.26	17.65	11.56	15.78	
0.2	9.38	15.7	9.81	15.12	
0.3	8.26	13.79	8.75	14.03	
0.4	6.03	11.52	6.34	11.09	
0.5	5.95	11.38	6.16	10.85	1-2 veh
0.6	5.5	11.13	5.81	10.69	
0.7	4.33	9.62	4.48	8.68	
0.8	4.1	9.34	4.5	8.97	
0.9	3.25	7.57	3.63	6.92	
1	2.95	7.15	3.35	6.48	

- The estimate error is 2-3 vehicles when MPR is 5%.
- The estimate error is 1-2 vehicles when MPR is 50%.

➢ Conclusion

This paper has studied the second-by-second queue profile estimation at signalized urban intersections based on EKF, only using limited data of CVs. Two challenges were overcome to design the queue estimator, which requires neither priori knowledge of the vehicle arrivals nor the market penetration rate (MPR) of CVs. The queue estimator was evaluated using simulation and field data of vehicle trajectories at under-saturated and oversaturated intersections under different levels of MPR of CVs. The evaluation results are satisfactory.

Acknowledgement: this study was supported by the Key Program project by Ministry of Science and Technology, China (project number: 2018YFB1600500), and in part by ZJU-UIUC Joint Research Centre Project No. DREMES202001, funded by Zhejiang University, and was led by Principal Supervisor Simon Hu. The real-world data used in this study comes from "pNEUMA – open-traffic.epfl.ch".

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